

A Sharp Threshold Atlas for Irreducible Network SIS: Global Endemicity and the Critical Perron Law

Route-A source-local certificate HCS-C271

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Abstract

We close the full phase portrait of the deterministic susceptible–infected–susceptible flow on an irreducible heterogeneous network. The spectral abscissa of the linearized Metzler matrix separates extinction from a unique endemic state. Above threshold, strict subhomogeneity gives global attraction, a Hurwitz endemic Jacobian, and componentwise transmission-rate monotonicity. At equality, normalized Perron vectors give the sharp vector law $tx(t) \rightarrow v/[\beta w^T \text{diag}(v)Av]$; 240 exact parameter rows and 720 equality samples audit every convention. No epidemic object is promoted to an arithmetic orbit or target determinant.

1 Frozen network flow

Let $A \geq 0$ be irreducible, let $D = \text{diag}(\delta_i)$ with $\delta_i > 0$, and let $\beta > 0$. On $Q = [0, 1]^n$ consider

$$\dot{x} = \beta \text{diag}(1 - x)Ax - Dx, \quad M = \beta A - D. \quad (1)$$

This is the physical ODE clock. Lajmanovich and Yorke established the classical nonhomogeneous deterministic epidemic lineage [1]; all normalizations below are stated locally.

At $x_i = 0$, the i th component of the vector field is nonnegative; at $x_i = 1$ it equals $-\delta_i$. Thus Q is invariant. Irreducibility and cooperativity make every nonzero orbit strictly positive for positive time.

Theorem 1 (threshold theorem). *If $s(M) < 0$, zero is globally exponentially stable. If $s(M) = 0$, zero is globally asymptotically stable. If $s(M) > 0$, (1) has exactly one $x_* \in (0, 1)^n$, and x_* attracts every nonzero point of Q .*

Proof. For $s = s(M) \leq 0$, choose $w \gg 0$ with $w^T M = sw^T$. Then

$$\frac{d}{dt} w^T x = sw^T x - \beta w^T \text{diag}(x)Ax.$$

This proves exponential decay for $s < 0$ and, by positivity and LaSalle’s argument, convergence to zero at equality. For $s > 0$, a small Perron vector is a strict subsolution and $\mathbf{1}$ is a strict supersolution. Existence follows by monotone iteration. On positive states,

$$f(\theta x) - \theta f(x) = \beta \theta (1 - \theta) \text{diag}(x)Ax \gg 0, \quad 0 < \theta < 1. \quad (2)$$

Largest- and smallest-ratio comparisons use (2) to prove uniqueness and to squeeze every positive orbit to x_* . $\square$

2 Endemic stability and monotonicity

At the endemic state,

$$J_* = \beta \text{diag}(1 - x_*)A - \beta \text{diag}(Ax_*) - D. \quad (3)$$

It is irreducible Metzler and $J_* x_* = -\beta \text{diag}(Ax_*)x_* \ll 0$; hence J_* is Hurwitz. Differentiating the equilibrium equation gives

$$\frac{dx_*}{d\beta} = -J_*^{-1} \text{diag}(1 - x_*)Ax_* \gg 0, \quad (4)$$

because $-J_*^{-1}$ is strictly positive. Equations (3)–(4) close both the local spectrum and the parameter direction, rather than stopping at an existence threshold.

3 The critical $1/t$ law

Assume $s(M) = 0$. Normalize $Mv = 0$, $w^T M = 0$, $v, w \gg 0$ by $w^T v = 1$, and set

$$\kappa = \beta w^T \text{diag}(v) A v > 0. \quad (5)$$

The zero eigenvalue is simple and the complementary spectrum is strictly stable. Put $P = v w^T$, $Q = I - P$, $E = \ker w^T$, and write $x = av + y$ with $a = w^T x > 0$ and $y = Qx \in E$. For $N(x) = -\beta \text{diag}(x) A x$,

$$\dot{a} = w^T N(x), \quad \dot{y} = M|_E y + QN(x). \quad (6)$$

Because $w \gg 0$ and $x \geq 0$, $a \asymp \|x\|_1$; hence $y = O(a)$ and $|\dot{a}| = O(a^2)$. The bounded quotient $z = y/a$ obeys

$$\dot{z} = M|_E z + QN(x)/a - (\dot{a}/a)z.$$

Critical extinction makes the last two terms tend to zero, while $M|_E$ is exponentially stable. Variation of constants gives $z \rightarrow 0$. Therefore

$$-\dot{a}/a^2 = \beta w^T \text{diag}(x/a) A (x/a) \rightarrow \kappa.$$

Thus $(1/a)' \rightarrow \kappa$, $ta \rightarrow 1/\kappa$, and $x/a = v + z \rightarrow v$. Consequently

$$tx(t) \rightarrow v/\kappa. \quad (7)$$

For an r -regular graph with uniform recovery δ , $v = \mathbf{1}$ and $w = \mathbf{1}/n$; equality means $\beta r = \delta$, so (7) becomes $tx(t) \rightarrow \mathbf{1}/\delta$. The invariant diagonal solves exactly $y' = -\delta y^2$.

4 Executable receipt and Route-A boundary

The canonical receipt contains 20 irreducible regular graphs, 240 exact threshold rows, and 720 rational critical samples. A producer-independent checker rebuilds connectivity, spectra, equilibria, Jacobian rates, and critical solutions; a separate symbolic reconstruction, fresh byte replay, and repaired-hash hostile suite close the release. Finite rows test formulas; the preceding argument proves the arbitrary-network theorem.

Under NO_BAD_EULER_OR_ROOT_NUMBER, the strict tuple is

$$(\text{AO_FAIL}, \text{A1_FAIL}, \text{A2_FAIL}, \text{A3_FAIL}, \text{A4_FAIL}),$$

overall ROUTE_A_REJECTED; Route B is disabled. A threshold, Perron vector, or endemic Jacobian is not a prime clock, primitive-orbit ledger, target divisor, or Hilbert–Pólya operator.

References

- [1] A. Lajmanovich and J. A. Yorke, *A deterministic model for gonorrhea in a nonhomogeneous population*, Mathematical Biosciences 28 (1976), 221–236, [doi:10.1016/0025-5564\(76\)90125-5](https://doi.org/10.1016/0025-5564(76)90125-5).